Flooring and Subfloor Build Process

As of: `2026-07-13` **Role:** canonical owner for the truck-bed floor stack, sealing, hardpoint registration, Lonseal installation, and post-install checks.

Owner-confirmed build decision

- The final floor-height limit is **`3/4 in` maximum**. The existing three removable `3/4 in` birch plywood panels are the final subfloor; do **not** add an underlayment layer.
- The stainless rivnuts are installed across mixed geometry: rib highs, valleys, and some rib-edge transitions. Active locations still need a hand-thread check, map, and square-load-path review during the plywood dry fit.
- One wide Gorilla Waterproof Patch & Seal Tape roll has been used. A second matching roll is planned for the remaining sealing work. In this protected, floored bed cavity, the tape is being used pragmatically as a well-adhered **dust/splash barrier**, not as an immersion-rated or structural repair.
- The priority is a flat, quiet, serviceable floor and hard-mounted modules—not chasing every preferred condition in a flooring-product brochure.

Current stack

Layer	Material	Working rule
Finish	Lonwood Madera Topseal, Oregano, one piece	Keep uncut/glue-free until hardpoint and dry-fit work is complete
Adhesive	Lonseal #650 two-part epoxy	Use the complete unit and follow its label/TDS for mix, spread, rolling, and cure
Subfloor	Three `3/4 in` birch plywood sections	Final height-locked substrate; bottom and edges sealed, top bond faces untreated
Insulation	`5/8 in` EPS in bed-rib valleys only	Filler/insulation; trim only where it makes plywood rock or conflicts with a hardpoint
Support	Truck-bed rib highs	The plywood must settle flat on its real bearing points

Practical floor-closeout rules

1. **Do not increase the floor height.** The current `3/4 in` floor is locked.
2. **Do not structurally fasten furniture only to plywood.** Use the bed rivnuts for the modules that need positive retention. Use plywood-only screws/cleats only where a panel edge or seam needs local flatness control.
3. **Do not carry bolt clamp load through EPS or an open valley.** Rib-high locations already have solid bed support. Valley locations need a measured rigid path between the plywood underside and the bed/rivnut bearing surface. Rib-edge locations need square bearing or they should be moved/retired.
4. **Do not bury unknown access needs.** Photograph and dimension installed hardpoint locations before the Lonseal dry fit. Use locator bolts, a template, or two bed datums so finished-floor holes can be recovered without guessing.
5. **Keep the bed clean and dry before floor closure.** Do not create a formal water-test project; just avoid obvious trapped water, loose debris, lifted tape, and known open dust paths.

Bed sealing approach

The goal is realistic: reduce dust, road splash, and nuisance intrusion under the floor.

- Continue using the Gorilla tape where it is cleanly adhered to the bed liner and is solving an observed seam/gap path.
- Hard-roll the tape, avoid trapped grit, and inspect corners/edges before the plywood goes back down.
- Keep tape, primer, and sealant clear of rivnut threads and the rivnut flange's structural bearing surface.
- Do not intentionally plug an obvious factory drain without a reason. This is a common-sense inspection item, not a requirement to map every body feature.
- After the first dusty or wet drive, inspect accessible floor edges and known former gap areas. Rework only actual failures.

Installed rivnuts and compression-sleeve routing

The rivnuts are now installed. Before floor closure:

1. Run a sacrificial bolt fully in and out of every active insert by hand; flag any spin, damaged thread, poor engagement, or bolt-axis problem.
2. Record each center and classify it as `rib high`, `valley`, or `rib edge/transition`.
3. Confirm the actual module tab/foot can approach the insert without forcing the bolt sideways.
4. Relocate or retire a rib-edge insert if its bolt axis or lower bearing surface cannot be made square. A short sealed filler bolt is acceptable for a retired insert after verifying that it engages without bottoming or protruding into a conflict.

The on-hand stainless sleeves are approximately `0.585 in OD x 7/16 in ID x 5/8 in long`. They are **not universal through-plywood compression limiters**: a `5/8 in` sleeve is `1/8 in` shorter than the `3/4 in` plywood before any valley gap is included.

Use them by geometry:

- **Rib high:** plywood already bears on bed metal. Do not place a `5/8 in` sleeve under the plywood and do not bore the full plywood thickness for it. Use the real module tab/foot or a broad top washer and tighten only enough to secure the joint without crushing the plywood.
- **Valley/EPS:** this is the useful location for the `5/8 in` sleeve, but only if the measured gap from the bed/rivnut bearing surface to the plywood underside is close to the sleeve's actual length. Clear the EPS around the sleeve, keep both sleeve ends square, and verify full lower bearing plus firm contact at the plywood underside. Replace or trim to measured length if the gap differs; do not pull the plywood down to meet a short sleeve or let a long sleeve prop it up.
- **Rib edge/transition:** do not let a square tube land on one edge or a slope. Move/retire the hardpoint or fabricate a properly seated local spacer/saddle only if that location is truly necessary.

If a future hardpoint needs a true sleeve through the plywood, make a custom limiter whose length equals the full distance from the module tab/washer underside to the lower metal bearing surface: `3/4 in plywood + measured under-plywood gap`. Bore only after the final hole map is proven. A `5/8 in` Forstner hole gives about `0.040 in` total clearance around the current `0.585 in` sleeve; the top module tab or washer must span that bore positively.

Floor and hardpoint sequence

1. Finish the map

- [ ] Finish only the remaining Desk/storage corrections needed to trust module feet/tabs.
- [ ] Photograph modules, shims, brackets, service openings, and the planned floor-tab locations.
- [ ] Record hardpoint centers from two truck-bed datums.
- [ ] Confirm the known rivnut locations are compatible with underside clearance and module access.

2. Finish sealing and verify installed rivnuts

- [ ] Complete the remaining practical tape/seal pass.
- [ ] Inspect tape adhesion and remove loose debris.
- [x] Install the stainless rivnuts.
- [ ] Hand-test/map every active insert; classify rib-high, valley, and rib-edge locations.
- [ ] Use the on-hand `5/8 in` sleeves only at measured valley/EPS gaps; move/retire angled locations that cannot seat square.
- [ ] Preserve the location map before covering the bed.

3. Settle the existing floor

- [ ] Inspect EPS; trim only proud, wet, damaged, or rocking pieces.
- [ ] Dry-fit the three plywood panels.
- [ ] Eliminate rocking, grit, sharp fasteners, unsupported edges, and visibly moving/high panel joints.
- [ ] Add only the local plywood retention needed to keep an edge or seam quiet and flat.
- [ ] Refit modules with temporary bolts and verify alignment, service access, and battery extraction.

4. Dry-fit and install Lonseal

- [ ] Acclimate and cut in shade on a clean, flat sacrificial surface near the installation temperature; Lonseal's vehicle guide identifies `65-85°F` as the optimal controlled range and recommends rough-cutting slightly large before final trimming.
- [ ] Before marking, verify the `72 x 96 in` sheet covers the fully assembled three-panel floor in the chosen grain direction. The `96 in` roll length has little nominal spare, so center front/rear coverage first.
- [ ] Use the existing fitted plywood panels as the primary full-size template: lay the Lonseal face-down, place the plywood bond faces down against the backing in exact truck order/seam spacing, mark `FRONT`, `REAR`, `DRIVER`, `PASSENGER`, and trace the exact outline.

- [ ] If a wheel-well or corner-pillar detail is not fully represented by the plywood edge, make a local kraft-paper/cardboard patch in the truck, label its `TOP` and two panel-edge/datum references, then flip it top-side-down onto the sheet backing with the plywood master. Do not assume left/right symmetry.
- [ ] Mark a second rough-cut line that leaves about `1/2-1 in` extra material everywhere practical. At a wheel-well/pillar opening, that means initially cutting the opening smaller so extra material remains on the `KEEP` side. Mark that side, use smooth radiused inside corners, and make multiple light knife passes over sacrificial backing.
- [ ] Move the rough-cut sheet into the truck unglued. Align centerline/front-rear references, make only the relief cuts needed to let it settle, and trim each wheel well/pillar in small increments. Finish with slight perimeter relief rather than jamming the sheet tightly against vertical metal.
- [ ] Preserve the hardpoint-hole recovery map and center marks, but do not punch/cut rivnut holes until the sheet's final position is registered.
- [ ] Confirm the plywood bond face is dry, smooth, dust-free, and untreated.
- [ ] Fill/sand only the joints, fastener depressions, or blemishes that would telegraph through the vinyl.
- [ ] Stage the correct trowel, roller method, mixer, blades, masking, cleanup supplies, protection, and a second person if useful.
- [ ] Follow the #650 product directions for complete A/B mixing, spread/open time, rolling, rerolling, and cure before heavy module reinstall.

5. Reinstall and prove

- [ ] Recover hardpoints from the map; do not blind-drill finished vinyl.
- [ ] Protect Lonseal during module installation.
- [ ] Install module bolts into the registered rivnuts and witness-mark the ones that matter.
- [ ] Confirm no module is loose, rubbing, or blocking electrical/plumbing/battery service access.
- [ ] Reinspect after the first local drive for loose fasteners, floor movement, vinyl indentation, or dust/moisture at known edges.

Simple go / no-go gates

Go when	No-go when
Tape/seal work is adhered and the bed is dry/clean	There is obvious loose tape, trapped debris, standing water, or a known open dust path
Rivnuts set firmly and pass a bolt-in/bolt-out check	An insert spins, has poor thread engagement, or cannot be serviced
Plywood lies flat and its seams do not move under foot	A panel rocks, a seam moves, or EPS holds the plywood high
All module locations are mapped and dry-refit works	Final module bolts or service access would be guesswork after vinyl
Adhesive tools, temperature, mixing, and cure time are ready	Rushing adhesive work would leave the floor unusable or force premature heavy loading

BOM synchronization

BOM owner: `bom/bom_estimated_items.csv`.

- `141`: `1/4-20` multi-grip rivet-nut tool set with stainless inserts — purchased; truck-bed inserts installed.
- `217`: `5/8 in` stainless spacer/compression sleeves — purchased; valley/EPS use only where the measured gap fits.
- `147`: `3/4 in` birch base plywood — purchased and height-locked.
- `148`: TotalBoat Halcyon edge/bottom sealer — purchased.
- `149`: Lonwood Madera Topseal sheet — purchased/in hand, unglued.
- `150`: Lonseal #650 `0.5 gal` — purchased/in hand, unmixed.
- `174`: exact-notch trowel — purchased.
- `175`: compact roller method — confirm physically before glue-down.
- `177`: polyurethane sealant — purchased/partially used.
- `178`: Gorilla Waterproof Patch & Seal Tape — two-roll/$28 scope; one used, one planned for the remaining practical bed-sealing pass.

Do not maintain a second hand-calculated flooring total here; the BOM CSV owns quantity, price, and purchase status.

References

- Lonseal Vehicle Installation Guide (repo copy: `references/lonseal flooring installation guide.pdf`).
- Lonseal #650 Technical Data Sheet: https://lonseal.com/wp-content/uploads/2024/11/TDS_650_110524.pdf.
- Gorilla Waterproof Patch & Seal Tape product page: https://gorillatough.com/product/gorilla-waterproofing-tape-black/.

Manufacturer documents guide product handling. This implementation file owns the build-specific decisions: the `3/4 in` height lock, practical tape/seal use, and stainless-rivnut hardpoint plan.